1. Does P-256's group order let the PKM coset factor base touch P-256's addition?

Date 2026-10-16 · **goal** GOAL-ECDLP-001 · **question** RQ-ECDLP-001 · **hypothesis** H-ECDLP-60de36 · **related** EXP-ECDLP-f49f75, —502ae2, —20a588 · **problem record** `problems/pkm-isogeny-chain-factor-base-no-fast-decomposition.md`

Answer. No. This is pen-and-paper work plus exact integer checks (`check_p256_structure.py`, output in `output.json`); it uses no solver and no timing. Labels: **derived** (proved below), **computed** (exact check on the published P-256 parameters, mine only), **recalled** (textbook fact, a pointer not support).

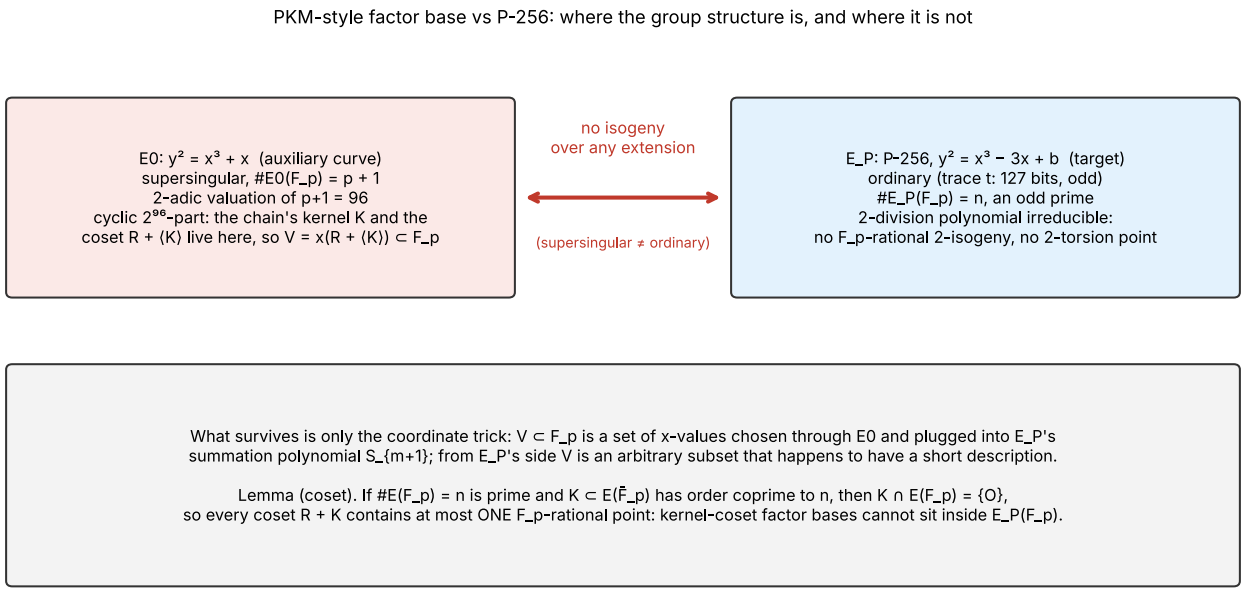


Figure 1: Where the group structure lives. Left: the auxiliary curve, whose cyclic 2-power subgroup the chain uses. Right: P-256. Nothing relates the two group laws.

1.1. Setting

The proposal takes $p = 2^{256} - 2^{224} + 2^{192} + 2^{96} - 1$ (P-256's field prime), notes $v_2(p+1) = 96$ (**computed**: 96), builds the chain Φ of 44 Vélu 2-isogenies on $E_0 : y^2 = x^3 + x$ (where $\#E_0(F_p) = p + 1$ since $p \equiv 3 \bmod 4$), and sets $V = x(R + \langle K \rangle) \subset F_p$, $|V| = 2^{44}$. It then uses V as a set of x-coordinates on E_P (P-256) and asks for $T = P_1 + \ldots + P_6$ with $x(P_i) \in V$.

1.2. Facts about P-256 (computed)

- The published G lies on the curve and $nG = O$ (so the constants are right); p and n are prime and n is odd.
- The trace $t = p + 1 - n$ has 127 bits and is odd; $t \neq 0$, so E_P is **ordinary**. E_0 is **supersingular** ($t = 0$).
- $x^3 - 3x + b$ is irreducible over F_p (the Frobenius orbit of x modulo it has size 3). So E_P has no point of order 2 over F_p and no F_p-rational 2-isogeny.
- For an odd prime l, an F_p-stable line in $E_{P[l]}$ (hence an F_p-rational l-isogeny) exists iff $t^2 - 4p$ is a square mod l (**recalled** criterion: Frobenius has characteristic polynomial $x^2 - tx + p$ on $E[l]$). It holds for $l = 3, 5, 11, 13, 17, 23, \ldots$ and fails for $l = 7, 19, 31, \ldots$ (full list in `output.json`).

1.3. Result 1: no map between the two group laws (derived)

An isogeny preserves supersingularity over $\overline{F}_p$, and every non-constant morphism of elliptic curves taking O to O is an isogeny (**recalled**). E_0 is supersingular and E_P is ordinary, so there is no non-constant morphism $E_0 \to E_P$ or back, over F_p or any extension. The cyclic subgroup $\langle K \rangle \subset E_0(F_p)$ therefore cannot be transported to E_P, and E_0's addition says nothing about E_P's.

1.4. Result 2: a kernel coset cannot sit inside $E_{P(F_p)}$ (derived)

Lemma. Let $\#E(F_p) = n$ be prime and let $K \subset E(\overline{F}_p)$ be a finite subgroup of order coprime to n. Then $K \cap E(F_p) = \{O\}$, so for every R, $|(R + K) \cap E(F_p)| \leq 1$.

Proof. If $R, R+k \in E(F_p)$ with $k \in K$, then $k = (R+k) - R \in E(F_p)$, so the order of k divides both $|K|$ and n, hence $k = O$. □

So even on E_P itself, a factor base made of an isogeny-kernel coset has at most one F_p-rational point, whatever isogeny degree is used. The F_p-rational l-isogenies for $l = 3, 5, 11, ...$ are real, but their kernel points live over extension fields. What survives of the proposal is only the **coordinate trick**: a subset $V \subset F_p$ chosen via E_0, whose elements are then used as x-values on E_P.

1.5. What this does and does not settle

- **Settled (derived + computed):** the 2-power torsion that makes $p+1$ special is a property of E_0 only. P-256′s group order n is prime and unrelated to $p+1$ beyond Hasse; nothing in n, t or p gives a structured F_p-subset of E_P that mirrors the chain.
- **Not settled:** whether the coordinate-defined set V could still make S_{m+1} restricted to V^m unusually easy. That is the empirical question the toy experiments address; they found no advantage in the encoding and, under block order, a roughly constant factor loss.
- **Not claimed:** anything about security beyond the toy scale; absence of an attack here is not security.

1.6. Next checks this suggests

1. State the "why is the chain slower, can it be fixed" question in the problem record (done in the same change) with the degree observation as the starting point.
2. If a new factor-base idea is proposed, apply the Lemma first: any construction that needs a rational **coset** inside $E_{P(F_p)}$ is ruled out before any experiment.